

HCI520: P1-LEARNING OBJECTIVES PROJECT

Project Objectives

For this project, I really wanted to push myself to move beyond the kind of intuition-based design I used a lot during my undergrad in UI. Back then, I would make design decisions mostly based on what looked good or felt right, but I didn't always have a structured way to explain why. Now, especially in HCI, I'm trying to be more intentional about how I think through problems.

The main goal here was to actually use Bloom's Revised Taxonomy in a practical way instead of just understanding it conceptually. I wanted to see how it works when you're forced to define learning outcomes clearly. One question I kept coming back to was how to tell the difference between someone just following steps versus someone who actually understands what they're doing. In this case, how do I know if a student is just setting up lights correctly versus actually understanding how light behaves and why certain setups work?

By the end of this, I wanted to get better at analyzing my own teaching goals with the same level of detail I usually apply when evaluating a user interface or user flow.

Method

I started by thinking about topics that connect both my background in design and my interest in how people think and learn. Photography felt like the most natural choice, especially tabletop photography, since it sits right at the intersection of visual design and technical understanding.

Once I chose the topic, I wrote out five learning objectives in plain, everyday language without worrying too much about taxonomy labels. After that, I went through each one and tried to map them to Bloom's categories. This part took more iteration than I expected. A lot of my original verbs were too vague or could fit into multiple categories.

What ended up helping me the most was focusing on the type of knowledge first. I would ask myself if I was trying to teach a fact, a concept, or a process. Once that became clearer, choosing the cognitive process (like understand vs apply vs analyze) felt more grounded and less like guessing.

TOPIC

Tabletop Photography: Mastering Light and Composition for Product Design

This lesson focuses on still life photography, specifically how controlled lighting and composition choices can communicate the quality and overall vibe of a product. The goal is to help students create images that feel intentional and professional, especially for portfolios or social media.

This is designed as a 3 hour studio-style session for design students or makers who need to document their work.

Prior Knowledge

Students are expected to already have:

- basic experience using a camera or smartphone (taking photos, adjusting focus)
- familiarity with simple visual principles like balance and contrast
- some understanding of where product photos are used, like Instagram or portfolios

Teaching Structure

Time	Activity	What's Happening	Why It Matters
Hour 1	Lighting demo	Introduction to three-point lighting and shadow behavior	Builds foundational understanding of how light works
Hour 2	Composition activity	Analyzing flat-lay vs eye-level shots and identifying patterns like rule of thirds	Helps students recognize visual structures
Hour 3	Shoot + critique	Students photograph an object and participate in peer critique	Moves from practice to reflection and evaluation

Results

Table of Learning Objectives

LO	English Description	Bloom's Category	Knowledge Type	Explanation
LO1	Identify components of a 3-point lighting setup	1.1 Recognizing	A. Factual	This is the starting point. Students need to visually recognize and label each light source correctly.
LO2	Predict how a bounce card changes shadows	2.2 Inferring	B. Conceptual	Students move beyond recall and start reasoning about how light behaves in different situations.
LO3	Execute a flat-lay composition using a grid	3.1 Executing	C. Procedural	This is the hands-on part where they apply a structured method to create a composition.
LO4	Differentiate between commercial and editorial lighting	4.1 Differentiating	B. Conceptual	Students analyze finished images and break down how lighting affects mood and style.
LO5	Critique a peer's photo using color harmony	5.2 Critiquing	C. Procedural	Students evaluate work using specific criteria instead of just giving general opinions.

Bloom's Taxonomy Grid

Knowledge \ Cognitive Process	1 Remember	2 Understand	3 Apply	4 Analyze	5 Evaluate	6 Create
A Factual	LO1					
B Conceptual		LO2		LO4		
C Procedural			LO3		LO5	
D Metacognitive						

Discussion
Problematic Points

One thing I found surprisingly difficult was separating apply from create. Initially, I assumed that taking a final photo would fall under create, but when I thought about the time constraint of a 3 hour session, it didn't really fit. Students are mostly applying existing rules like lighting setups and composition grids rather than developing something completely original. That made me rethink how I define creativity in a structured learning setting.

Reflection

The translation process definitely required more interpretation than I expected. For example, something like critiquing a photo sounds straightforward, but when I tried to map it properly, I realized it only works if students actually have clear criteria to base their judgment on. Without that, it becomes more like giving opinions rather than evaluating.

Another thing I noticed is that most of my objectives fall under conceptual and procedural knowledge, with only one being purely factual. I think this reflects how I naturally approach design. I tend to focus more on understanding systems and applying them rather than memorizing isolated facts.

Assessment

The objectives feel pretty clear in terms of how they can be assessed.

- LO1 could be tested using a simple visual labeling activity
- LO3 can be assessed by checking if the composition follows the grid correctly
- LO5 works well as a structured peer critique where students have to explain their reasoning in a few sentences

Conclusion

This project honestly changed how I think about both teaching and design. Earlier, I focused mostly on the final output, like whether the photo looked good. Now I'm paying more attention to what's happening cognitively while someone is learning or doing a task.

If I were to build on this, I would definitely include something in the metacognitive category. For example, asking students to reflect on how their understanding of lighting or composition changed during the session. That feels like a missing piece right now.

Overall, this framework feels useful beyond just teaching. I can see myself applying this kind of structured thinking when presenting UX research or even when breaking down how users learn a system over time.